

Towards Retrieval-Augmented Large Language Models: Data Management and System Design

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Abstract—Retrieval-augmented generation (RAG) has become a transformative approach for enhancing large language models (LLMs) by integrating external, reliable, and up-to-date knowledge. This addresses critical limitations such as hallucinations and outdated internal information. This tutorial delves into the evolution and frameworks of RAG, emphasizing the pivotal role of data management technologies in optimizing query processing, storage, indexing, and efficiency. It explores how RAG systems can deliver high-quality, context-aware outputs through efficient retrieval and integration, covering key topics such as retrieval-augmented LLM (RA-LLM) architectures, retrieval techniques, learning methodologies, and applications in NLP and domain-specific tasks. Challenges like customized query and generation, real-time retrieval, and trustworthy RAG are discussed alongside future directions and opportunities for innovation. Designed for students, researchers, and industry practitioners with basic artificial intelligence and data engineering knowledge, this tutorial offers practical insights into designing data management-powered RAG systems. It inspires the exploration of novel solutions in this rapidly evolving field.

Index Terms—Retrieval-Augmented Generation, Data management, Large Language Model

I. INTRODUCTION

Retrieval-augmented generation (RAG) is a pivotal technique in AI, enhancing the capabilities of Large Language Models (LLMs) by integrating external, up-to-date knowledge into their workflows. LLMs, such as GPT [1] and BERT [2], have revolutionized natural language understanding and generation but suffer limitations like hallucinations, domain knowledge gaps, and the need for costly fine-tuning [3]. RAG bridges these gaps by leveraging retrieval mechanisms to access relevant external databases during generation, thus improving accuracy and relevance [4]. It has proven instrumental in knowledge-intensive tasks like answering questions, chatbots, and recommender systems and mitigating issues like hallucinations in legal [5] and medical queries [6]. For instance, retrieval-enhanced systems [7], [8] integrate retrieved information with input queries to refine response generation, while

retrieval-augmented frameworks [9], [10] dynamically access external knowledge bases for domain-specific applications. Recent advancements also include self-reflective retrieval systems [11] that assess the utility of retrieved content and dense retrieval frameworks [12] that enhance attention mechanisms. These innovations underscore the evolving synergy between retrieval and generation, showcasing RAG as a cornerstone of AI-driven applications.

Data management technologies significantly enhance RAG systems by improving query processing, storage, indexing, and execution, making them an important topic to explore for both research and application value. Efficient similarity-based search methods optimize query performance for open-domain tasks [13]–[15]. Advanced storage and indexing techniques enable scalable handling of high-dimensional embeddings, supporting large-scale nearest neighbor search capabilities [16], [17]. Query optimization frameworks improve system efficiency by balancing latency and quality through pipeline management and adaptive retrieval intervals [18]–[20]. Furthermore, knowledge management modules integrate domain-specific augmentation by leveraging vector databases, supporting both structured and unstructured data queries through RAG-enhanced LLMs [21].

To this end, we propose to present the tutorial on the targeted topic to share our knowledge and insights with practitioners from both the academia and industry communities who are interested in this topic and try to apply in their own projects. Our tutor team consists of well-experienced researchers with expertise relevant to the topic, which will be introduced in the following Sections (Sec. II–Sec. IV). We plan to organize the tutorial in six major parts, which is detailed in Section VI.

II. TUTORIAL PRESENTERS

Six tutors will contribute to this tutorial. Their brief information is as follows.

III. TUTOR INFORMATION

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IV. BRIEF BIOS

In this section, we will give brief bios for all presenters.

- **Dr. Wenqi Fan** is an assistant professor of the Department of Computing at The Hong Kong Polytechnic University (PolyU). He received his Ph.D. degree from the City University of Hong Kong (CityU) in 2020. From 2018 to 2020, he was a visiting research scholar at Michigan State University. His research interests are machine learning and data mining, focusing on Recommender Systems, Graph Neural Networks, and Large Language Models (LLMs). He has published innovative papers in top-tier journals and conferences such as IEEE TKDE, KDD, WWW, NeurIPS, ICDE, IJCAI, AAAI, RecSys, WSDM, and SDM. He serves as a top-tier conference (senior) program committee member (e.g., ICML, ICLR, NeurIPS, KDD, WWW, AAAI, IJCAI, CIKM, WSDM) and journal reviewers (e.g., TKDE, TIST, TKDD, TOIS, TAI). He also serves as the lead tutor of tutorials in top-tier conferences (e.g., WSDM 2023, WWW 2021/2022, IJCAI 2021, and ICAPS 2021). More information about him can be found at <https://wenqifan03.github.io/>.
- **Pangjing Wu** is currently a PhD student at The Hong Kong Polytechnic University, supervised by Dr. Wenqi Fan and Prof. Qing Li. He got his BSc and M.Eng. degree at Hohai University (HHU). His research interests include data integrity, data pricing, large language models (LLMs), and reinforcement learning. More information about him can be found at <https://pangjing-wu.github.io/>.
- **Dr. Yujuan Ding** is a research assistant professor at school of fashion and textile, The Hong Kong Polytechnic University (PolyU). She received her Ph.D. degree from the same university in 2021. From 2019 to 2020, she visited the school of Computing at National University of Singapore as a research student. She has extensive research experience in the field of multimedia analysis, large language models (LLMs), and information retrieval across various domains, specifically focusing on the topics of recommender systems, time-series modeling,

vision-language modeling and other AI applications. She has published her research outcome in top-tier journals and conferences including ACM CSUR, IEEE TMM, IEEE TSCVT, ACM MM, ICCV, etc. She was the recipient of the best student paper award of ICMR 2021. She serves as the program committee members for top-tier conferences (e.g., AAAI, ACL, ACM MM, EMNLP) and reviewers for journals. More information about her can be found at <https://joanding.github.io/>.

- **Liangbo Ning** is currently a PhD student at The Hong Kong Polytechnic University, supervised by Prof. Qing Li and Dr. Wenqi Fan. He got his B.Eng. and MSc degrees at Northwestern Polytechnical University (NPU). His research interests include adversarial attacks, large language models (LLMs), and pattern recognition. More information about him can be found at <https://biglemon-ning.github.io/>.
- **Shijie Wang** is currently a PhD student at The Hong Kong Polytechnic University, supervised by Dr. Wenqi Fan. He got his BSc degree at the University of Liverpool (Uol). His research interests include recommender systems, large language models (LLMs), and graph neural networks. More information about him can be found at <https://sjay-wang.github.io/>.
- **Prof. Qing Li** is currently a Chair Professor (data science) and the Head of the Department of Computing, The Hong Kong Polytechnic University. He received the B.Eng. degree from Hunan University, Changsha, China, the M.Sc. and Ph.D. degrees from the University of Southern California, Los Angeles, all in computer science. His research interests include object modeling, multimedia databases, social media, and recommender systems. He is a Fellow of IEEE and IET, a member of ACM SIGMOD and IEEE Technical Committee on Data Engineering. He is the chairperson of the Hong Kong Web Society and is a steering committee member of DASFAA, ICWL, and WISE Society. More information about him can be found at <https://www4.comp.polyu.edu.hk/~csqli/>.

V. TUTORIAL LENGTH

We plan to make a 3-hour length tutorial at ICDE 2025.

VI. OUTLINE OF THE TUTORIAL

This tutorial is expected to be presented as a half-day lecture with slides and handouts to the audience. The arrangement of detailed content to be introduced in this tutorial is given as follows:

- Introduction (10 mins)
 - Background of the topic, basic concept and ideas of LLM, RAG and data management
 - The tutorial organization
- Architecture of Combining RAG with LLMs (40 mins)
 - RA-LLM architecture overview
 - Retriever in RAG-LLMs

- Retrieval results integration
- Pre/Post-retrieval techniques
- Special RA-LLM paradigms
- Data Management for RAG (40 mins)
 - Preliminaries of Data Management
 - RAG-powered by Data Management
 - Popular Vector Database Management Systems
- Learning of RAG (30 mins)
 - Training-free methods
 - Training-based methods
- Applications of RAG (30 mins)
 - NLP application
 - Downstream tasks
 - Domain-specific applications
- Future Directions of Data Management and RAG (20 mins)
 - Customized Query and Generation
 - Challenges in Real-Time Retrieval
 - Trustworthy RAG
- QA session (10 mins)

VII. POTENTIAL TARGET AUDIENCE

The audience of this tutorial could be college students, researchers in academic institutions, and industrial database and AI labs interested in data engineering, artificial intelligence, and the latest trends in LLMs. Audiences are expected to have some basic understanding of data engineering, artificial intelligence, information retrieval, and natural language processing. However, the tutorial will be presented at the college junior/senior level so that it can be comfortably followed by academic researchers or industrial practitioners who are interested in this emerging field. After attending this tutorial, the audience is expected to comprehensively understand data management in retrieval augmented generation and learn how to design data management-powered RAG for real-world applications. It is also hopeful that the audience gets inspired by this tutorial and can step further in this field to explore novel opportunities in data management-powered RAG.

VIII. HANDS-ON TUTORIAL

This tutorial will primarily consist of a presentation delivered through detailed slides to introduce and explain the related technology. It is not a hands-on session; attendees will not need to perform tasks during the tutorial.

IX. PRIOR OFFERINGS

Our tutorial will partly overlap with one tutorial made in KDD'24 presented, titled *RAG Meets LLMs: Towards Retrieval-Augmented Large Language Models*¹ [22] on August 25, 2024. It comprehensively reviews existing research studies in retrieval-augmented large language models (RA-LLMs), covering three primary technical perspectives: architectures, training strategies, and applications. This tutorial and ours

cover two main technical parts of RAG: the model architecture and learning strategy. We believe these two aspects are essential to learning in technical when teaching about RAG. However, instead of introducing RAG for LLMs, we pay special attention to how data management technologies power retrieval-augmented generation. More specifically, we will introduce several key methods highlighting the role of data management technologies in enhancing retrieval-augmented generation as well as challenges [23]. These include large-scale vector search [16] and advanced indexing techniques, such as graph-based similarity search [24]–[26], which facilitate efficient nearest-neighbor searches in large-scale datasets. Dynamic embedding updates address embedding drift, ensuring that the data storage remain relevant over time [27]. Furthermore, data management-powered frameworks enable real-time query execution, crucial for applications requiring immediate responses [20], [28]. These advancements underscore the essential role of data management in scaling RAG systems and delivering high-quality, contextually enriched outputs.

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¹<https://advanced-recommender-systems.github.io/RAG-Meets-LLMs/>

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